ECSCI24302 – Machine Learning Essentials

PROJECT-BASED LEARNING (PBL)

Course: ECSCI24302 – Machine Learning Essentials

Project Instructions

1. Project Title

End-to-End Machine Learning Project: From Data Exploration to Model Evaluation

2. Project Objective

The objective of this project is to enable students to apply the complete Machine Learning workflow to a real-world dataset. Students will identify a suitable problem, select an appropriate dataset from a recognized public repository, analyse and preprocess the data, develop suitable Machine Learning models, evaluate and compare their performance, and communicate their findings.

The project emphasizes learning by doing and requires students to understand the complete ML pipeline rather than applying an algorithm in isolation.

3. Learning Outcomes

- Formulate a clear real-world problem suitable for Machine Learning.
- Select and understand a suitable dataset from Kaggle or the UCI Machine Learning Repository.
- Perform data cleaning, exploratory data analysis, preprocessing, and feature engineering.
- Select appropriate Machine Learning algorithms based on the problem and data.
- Train, validate, tune, and evaluate Machine Learning models.
- Compare models using appropriate performance metrics.
- Interpret results, analyse errors, and justify modelling decisions.
- Document and present the complete Machine Learning workflow professionally.

4. Project Team and Constraints

- The project must be completed in a team of **2–3 students**.
- A team cannot have more than 3 members (Inter-batch project teams are not permitted).
- Each team must work on a clearly defined Machine Learning problem.
- Each team must use a dataset obtained from either Kaggle or the UCI Machine Learning Repository.
- The selected dataset must be sufficiently meaningful for applying an ML pipeline; very small toy datasets should not be selected.
- Teams must finalize both their team members and dataset by 30 September 2026.
- Duplicate datasets/topics across teams should be avoided. Dataset approval will be based on suitability and availability.
- Students must cite the original dataset source in the final report.

5. Dataset Selection

Each team must select one dataset from either **Kaggle or the UCI Machine Learning Repository**. The dataset should support a meaningful Machine Learning problem and should provide enough observations and features for analysis, modelling, and evaluation.

The project may involve classification, regression, clustering, or another appropriate Machine Learning task. The selected task must be justified based on the problem and characteristics of the dataset.

Students should first identify the real-world problem and then select an appropriate dataset and ML approach.

6. Suggested Project Domains

- Healthcare and disease prediction
- Finance and credit-risk analysis
- Customer behaviour and churn prediction
- Education and student performance
- Retail and sales prediction
- Transportation and traffic analysis
- Environment and weather
- Sports analytics
- Text classification
- Audio or speech classification
- Cybersecurity and anomaly detection
- Energy consumption prediction
- Any other suitable domain approved by the instructor.

The above domains are illustrative. Teams may propose another domain if the problem and dataset are suitable.

7. Complete Machine Learning Pipeline

Every project must demonstrate the following stages. The exact techniques will depend on the selected dataset and problem.

Stage	Required Work
1. Problem Definition	Define the real-world problem, objective, inputs, target/output where applicable, and expected outcome.
2. Dataset Acquisition	Obtain the dataset from Kaggle or UCI and document dataset name, source, link, size, and relevant usage/license information.
3. Dataset Understanding	Describe observations, features, target, data types, categorical/numerical variables, and identifier columns.
4. Data Quality Check	Check missing values, duplicate records, inconsistent values, invalid entries, data types, and potential outliers.
5. Exploratory Data Analysis	Perform univariate and bivariate analysis using suitable statistics and visualizations. Identify

	important patterns and relationships.
6. Data Preprocessing	Handle missing values, duplicates, outliers, categorical variables, and other data-quality issues with justification.
7. Feature Engineering	Create, transform, combine, encode, scale, or otherwise modify features where appropriate. Explain each decision.
8. Feature Selection	Identify relevant features using appropriate statistical, correlation-based, model-based, or other justified techniques.
9. Train-Test Split	Divide data into training and testing sets. Use validation data or cross-validation during model development when appropriate.
10. Baseline Model	Build a simple baseline to provide a reference for more advanced models.
11. Model Selection	Select at least three suitable ML algorithms and explain why they are appropriate.
12. Model Training	Train the selected models using the training data.
13. Hyperparameter Tuning	Tune important hyperparameters using validation or cross-validation.
14. Model Evaluation	Evaluate final models on the untouched test set using suitable metrics.
15. Model Comparison	Compare models in a single table and discuss performance, complexity, and practical considerations.
16. Error Analysis	Investigate incorrect predictions or large errors and identify possible reasons.
17. Interpretation	Explain important features, model behaviour, and meaningful patterns found in the results.
18. Final Prediction / Demonstration	Demonstrate the final model on new/unseen input or provide an appropriate project demonstration.
19. Conclusion	Summarize the methodology, findings, limitations, and possible future improvements.

8. Minimum Technical Requirements

- Perform EDA using appropriate plots and statistical summaries.
- Handle missing values and explain the selected strategy.
- Identify and discuss outliers where relevant.
- Perform appropriate encoding for categorical variables.
- Apply feature scaling when required by selected algorithms.
- Use at least three suitable Machine Learning models.
- Include a baseline model.
- Perform hyperparameter tuning for at least two selected models where meaningful hyperparameters exist.
- Use cross-validation where appropriate during model development.
- Use evaluation metrics suitable for the problem.
- Prepare a final model-comparison table.
- Perform error analysis and discuss limitations.

9. Choosing Evaluation Metrics

The evaluation metrics must match the type of Machine Learning problem.

Problem Type	Possible Metrics
Classification	Accuracy, Precision, Recall, F1-score, ROC-AUC, Confusion Matrix
Regression	MAE, MSE, RMSE, R^2
Clustering	Silhouette Score and appropriate cluster-analysis measures
Other problems	Use metrics appropriate to the specific problem and justify the choice.

Students must explain why the selected metrics are appropriate for their problem.

10. Timeline and Milestones

Phase	Timeline	Expected Deliverable
Phase 1: Team & Dataset Finalization	17–30 September 2026	Team members, project title, problem statement, dataset source/link, and brief justification.
Phase 2: Problem & Dataset Study	1–7 October 2026	Problem definition, dataset description, feature/target identification, and initial data-quality analysis.
Phase 3: EDA & Preprocessing	8–17 October 2026	EDA plots, observations, missing-value treatment, outlier analysis, encoding, scaling, and cleaned data/pipeline.
Phase 4: Feature Engineering & Baseline	18–23 October 2026	Engineered/selected features and baseline model with initial evaluation.
Phase 5: Model Development & Tuning	24–31 October 2026	At least three ML models, hyperparameter tuning, cross-validation where appropriate, and experiment records.
Phase 6: Final Evaluation & Documentation	1–7 November 2026	Final evaluation, model comparison, error analysis, report, presentation, and demonstration.

Mandatory deadlines: Team and dataset finalization by 30 September 2026. Complete project submission by 7 November 2026 (first week of November).

11. Project Proposal – Due 30 September 2026

1. Team member names and enrollment numbers.
2. Project title.
3. Problem statement.
4. Motivation and real-world relevance.

5. Dataset name and source (Kaggle or UCI).
6. Dataset link.
7. Number of observations and features.
8. Target variable and its type, if applicable.
9. Expected ML task: classification, regression, clustering, etc.
10. Proposed Machine Learning models.
11. Expected evaluation metrics.

12. Final Project Report Structure

Title Page
Abstract / Project Summary
1. Introduction
2. Problem Statement and Motivation
3. Dataset Description and Source
4. Data Understanding and EDA
5. Data Preprocessing
6. Feature Engineering and Feature Selection
7. Machine Learning Methodology
8. Baseline Model
9. Model Development
10. Hyperparameter Tuning
11. Results and Model Comparison
12. Error Analysis
13. Model Interpretation
14. Limitations
15. Conclusion and Future Work
References

13. Results to be Reported

- Dataset summary table.
- Missing-value and duplicate analysis.
- Important EDA visualizations with observations.
- Preprocessing decisions and justification.
- Feature engineering/selection details.
- Baseline model results.
- Results of at least three ML models.
- Hyperparameter search space and selected parameters.
- Cross-validation results where applicable.
- Final test-set performance.
- Appropriate prediction/error plots.
- Final model comparison table.
- Error analysis.
- Feature importance/model interpretation where applicable.

14. Model Comparison Table

Model	Best Hyperparameters	Validation Result	Test Metric(s)	Observations
Baseline				
Model 1				
Model 2				
Model 3				
Additional Model (if used)				

The final selected model must be justified using experimental evidence. Students should consider performance as well as model complexity, interpretability, computational requirements, and suitability for the problem.

15. Data Leakage and Experimental Practice

Students must ensure that information from the test set does not influence model training, preprocessing decisions, feature selection, or hyperparameter tuning.

- Fit data-dependent preprocessing steps using training data only.
- Do not use the test set to select hyperparameters.
- Do not repeatedly inspect test performance and modify the model based on it.
- Use pipelines where appropriate.
- Identify and document features that may contain target or future information.

16. Tools and Technologies

- Python
- NumPy
- Pandas
- Matplotlib / Seaborn
- Scikit-learn
- Jupyter Notebook / Google Colab
- Other libraries only when necessary and properly documented.

17. Academic Integrity

- All dataset sources must be clearly cited.
- Code obtained or adapted from external sources must be acknowledged.
- Plagiarism in code, report, analysis, or presentation is not permitted.
- Students must understand and be able to explain all code used in their project.
- Each team member must be able to explain their contribution and the complete project pipeline during the viva.

18. Assessment Scheme

Component	Weightage
Problem formulation & dataset selection	10%

EDA & data preprocessing	20%
Feature engineering & selection	10%
Model development & implementation	20%
Hyperparameter tuning & evaluation	15%
Analysis, interpretation & model comparison	10%
Final report & presentation/demo	10%
Viva voce	5%
Total	100%

19. Viva Voce – Areas of Evaluation

- Understanding of the selected dataset.
- Reason for selecting the problem and dataset.
- Understanding of preprocessing steps.
- Reason for selecting features.
- Understanding of selected algorithms.
- Understanding of hyperparameters.
- Interpretation of evaluation metrics.
- Ability to explain model comparison.
- Understanding of overfitting, underfitting, and data leakage.
- Ability to explain the team's own code and results.

20. Final Presentation / Demonstration

- Problem and motivation
- Dataset and EDA
- Preprocessing and feature engineering
- Models developed
- Hyperparameter tuning
- Evaluation and comparison
- Error analysis
- Final model and prediction/demo
- Limitations and future work

21. Final Submission Checklist

- Team of 2–3 members finalized.
- Dataset and project finalized by 30 September 2026.
- Dataset source and link documented.
- Problem statement clearly defined.
- Complete ML pipeline demonstrated.
- EDA completed with meaningful observations.
- Data preprocessing documented.
- Feature engineering/selection documented.
- At least three suitable ML models trained.
- Hyperparameter tuning performed where applicable.
- Final evaluation performed on unseen test data.

- Model comparison table included.
- Error analysis included.
- Final report completed.
- Code/notebook organized and reproducible.
- Presentation and demonstration prepared.
- Project completed and submitted by 7 November 2026.